

Lab 1 – Setting Up Flutter and Running Your First App

1. Objective

This lab helps students set up the Flutter environment and run their first Flutter application. Students will learn how to install tools, create a project, execute it on a device/emulator, and perform basic UI customization.

2. Requirements

Students must:

- Install Flutter SDK
- Configure IDE (Android Studio or VS Code)
- Create and run a Flutter app
- Customize the layout
- Submit screenshots as evidence

3. Guided Steps

Exercise 1 – Flutter Environment Setup

Goal: Install and verify Flutter tools.

Steps

1. Install Flutter SDK.
2. Add Flutter to PATH.
3. Run:

`flutter doctor`

4. Ensure all items are properly configured.
5. Verify Flutter & Dart plugins in Android Studio.

Deliverables

- Screenshot of flutter doctor

```

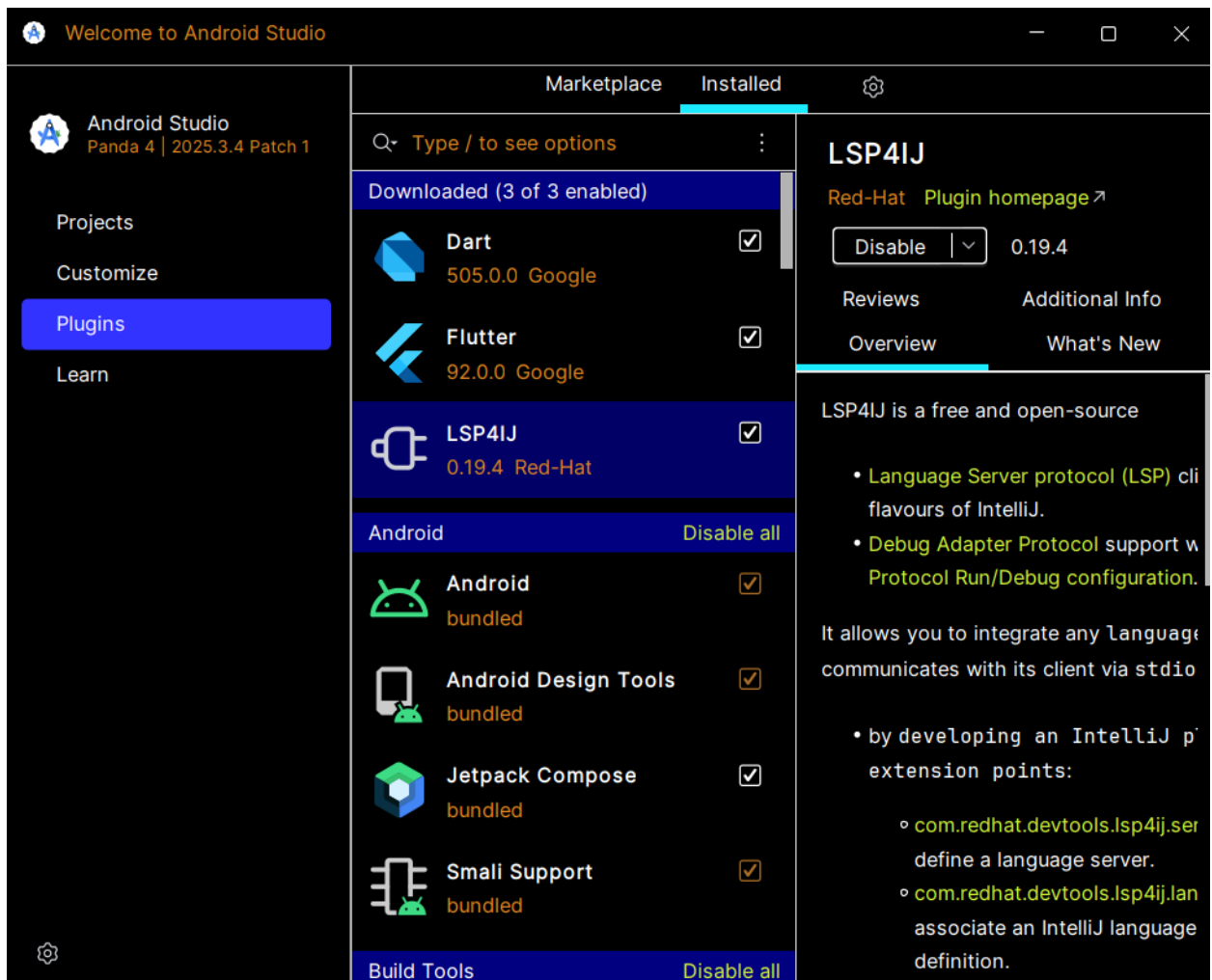
D:\flutter_windows_3.44.0-stable\flutter\bin>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.44.0, on Microsoft Windows [Version 10.0.26200.8457], locale en-US)
[✓] Windows Version (Windows 11 or higher, 25H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
[✓] Chrome - develop for the web
[X] Visual Studio - develop Windows apps
    X Visual Studio not installed; this is necessary to develop Windows apps.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its default components
[✓] Connected device (3 available)
[✓] Network resources

! Doctor found issues in 1 category.

D:\flutter_windows_3.44.0-stable\flutter\bin>

```

-
- Screenshot of Flutter/Dart plugin screen



Exercise 2 – Create and Run Your First Flutter App

Goal: Learn how to create a Flutter project and run the default app.

Steps

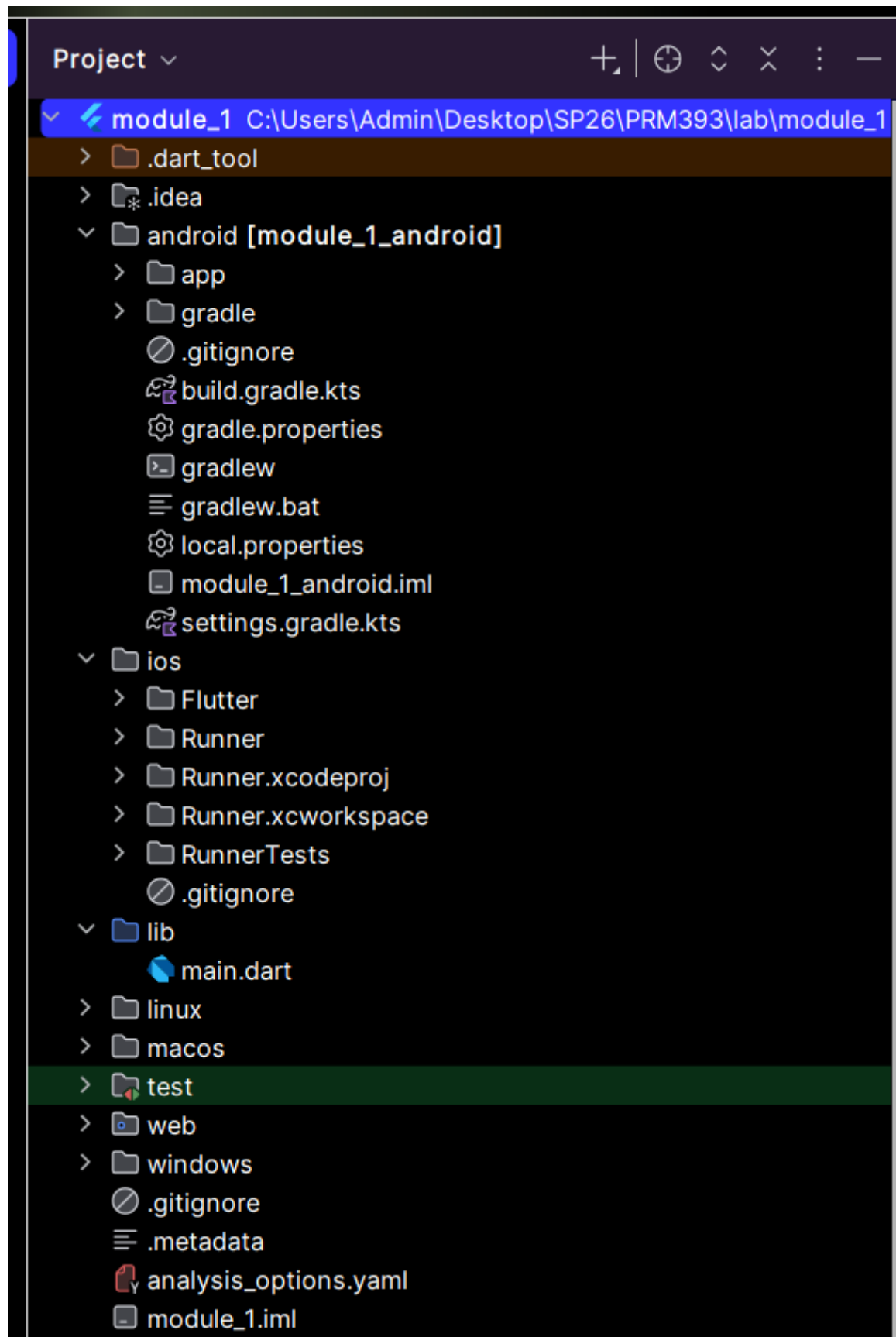
1. Open Android Studio → **New Flutter Project**
2. Use project name: `hello_flutter_lab1`
3. Explore key folders: `lib/`, `android/`, `ios/`, `pubspec.yaml`
4. Start emulator or connect device
5. Run the default counter app
6. Modify the AppBar title in `main.dart`:

title: 'My First Flutter App',

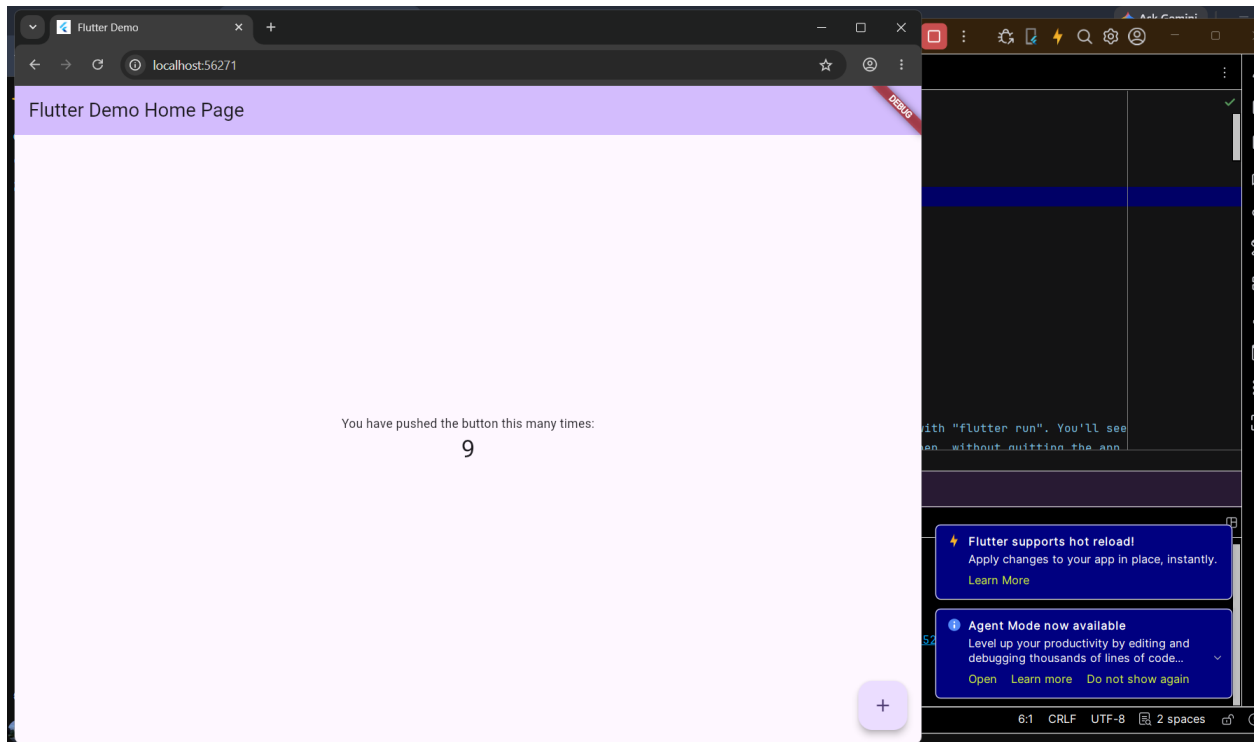
7. Use **Hot Reload** and observe changes

Deliverables

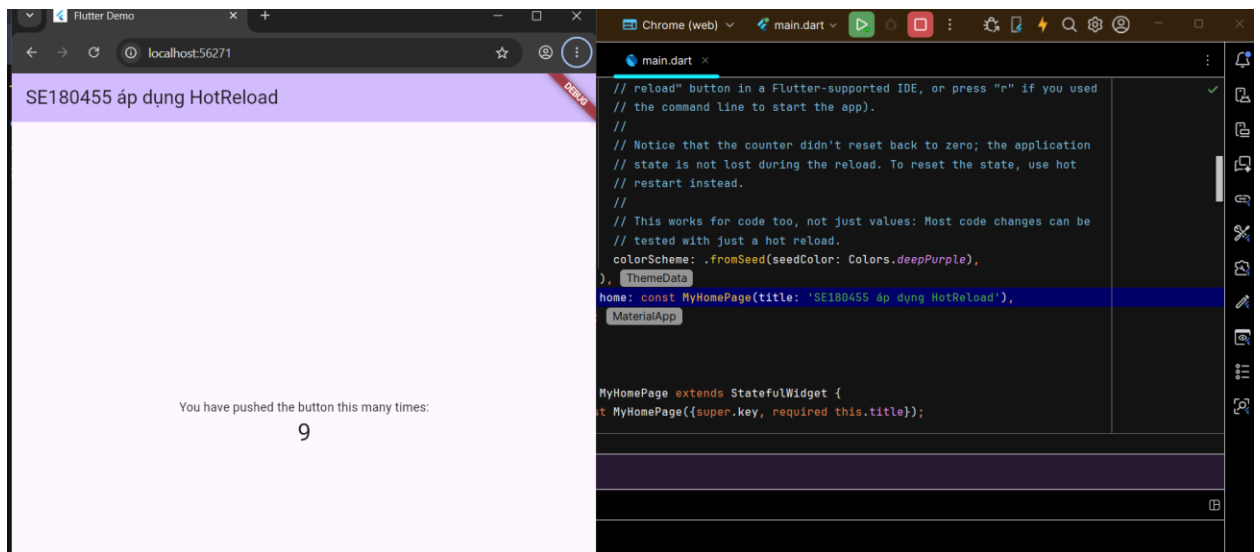
- Screenshot of project structure



- Screenshot of running counter app



- Screenshot showing updated title via Hot Reload



Exercise 3 – Customize Your First Flutter UI

Goal: Build a simple custom UI with widgets learned in class.

Steps

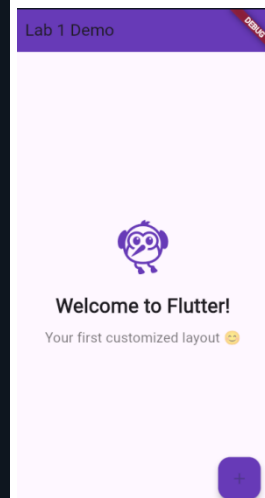
1. Replace default code in main.dart with:

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Lab 1 Demo'),
          backgroundColor: Colors.deepPurple,
        ),
        body: const Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Icon(Icons.flutter_dash, size: 80, color: Colors.deepPurple),
              SizedBox(height: 20),
              Text(
                'Welcome to Flutter!',
                style: TextStyle(fontSize: 26, fontWeight: FontWeight.bold),
              ),
              SizedBox(height: 10),
              Text(
                'Your first customized layout 😊',
                style: TextStyle(fontSize: 18, color: Colors.black54),
              ),
            ],
          ),
        ),
        floatingActionButton: FloatingActionButton(
          onPressed: null,
          backgroundColor: Colors.deepPurple,
          child: Icon(Icons.add),
        ),
      ),
    );
  }
}
```



2. Perform Hot Reload and verify UI updates.
3. Modify colors/text and reload again.

Exercise 4 – Reflection Questions

1. What is the purpose of the flutter doctor command?

- Mục đích của lệnh flutter doctor : Kiểm tra flutter đã cài đặt đúng chưa,
- Android Studio / VS Code đã được cấu hình hay chưa.

- Android SDK hoặc XCode có bị thiếu hay không
- Có lỗi nào cần sửa trước khi chạy Flutter không, kiểm tra được lỗi môi trường.

2. What file acts as the entry point of a Flutter application?

File nào là điểm bắt đầu của Flutter

- main.dart : Trong file chứa hàm main().

3. Explain the difference between Hot Reload and Hot Restart.

Điểm khác biệt giữa Hot Reload và Hot Restart:

Hot Reload	Hot Restart
Cập nhật giao diện nhanh mà không mất state	Khởi động lại ứng dụng hoàn toàn
Giữ nguyên dữ liệu hiện tại	Mất toàn bộ state
Nhanh hơn	Chậm hơn một chút
Dùng khi chỉnh sửa UI	Dùng khi thay đổi logic lớn hoặc biến global

4. How does runApp() build the widget tree?

Cấu trúc xây dựng như nào:

- Khi chạy runApp() là điểm bắt đầu để hiển thị giao diện Flutter
- Flutter sẽ:
 - + Lấy widget gốc (MyApp)
 - + Xây dựng cây widget
 - + Render giao diện lên màn hình
 - + Theo dõi sự thay đổi để cập nhật UI khi cần
- Mỗi widget trong Flutter sẽ chứa các widget con, tạo thành cấu trúc cây widget

5. Describe how Flutter's architecture enables cross-platform development.

Cách kiến trúc Flutter phát triển đa nền tảng:

- Sử dụng 1 codebase duy nhất cho Android, IOS, Web, Desktop
- Flutter sử dụng ngôn ngữ Dart
- Có rendering riêng là Skia, giúp vẽ giao diện trực tiếp thay vì dùng Native UI Components
- Giúp giao diện được đồng bộ trên nhiều nền tảng

Ưu điểm :

- Tiết kiệm thời gian phát triển
- Dễ bảo trì
- Hiệu năng cao
- UI đồng nhất trên các nền tảng

4. Expected Results

- Successful setup of Flutter environment
- Ability to create and run Flutter projects
- Understanding of widgets, layout, and hot reload
- Complete screenshots for all required steps

5. Submission

- Zipped Flutter project folder or GitHub link
- All screenshots
- Written answers for reflection questions